Q1. (Medium) Geostationary satellite characteristic?

A. Low altitude

B. 24 hr orbit

C. Polar orbit

D. Random motion

Correct Answer: 2

Explanation: Geostationary satellites orbit the Earth in 24 hours and remain fixed relative to a point on Earth.

Q2. (Medium) Circuit switching establishes?

A. Packet route

B. Dedicated path

C. Shared channel

D. Virtual path

Correct Answer: 2

Explanation: Circuit switching creates a dedicated communication path before data transmission begins.

Q3. (Medium) Main drawback of circuit switching?

A. Packet drop

B. High loss

C. No bandwidth

D. Setup delay

Correct Answer: 4

Explanation: It takes time to establish a dedicated path, causing setup delay.

Q4. (Medium) Satellite communication uses?

A. Infrared

B. Twisted pair

C. Fiber only

D. Microwaves

Correct Answer: 4

Explanation: Satellite communication uses microwaves as they can travel long distances in space.

Q5. (Medium) Switched network example?

A. Scanner

B. Standalone PC

C. Printer

D. Telephone network

Correct Answer: 4

Explanation: Telephone networks are classic examples of switched networks.

Q6. (Medium) Full duplex means?

A. Two-way simultaneous

B. One-way only

C. Half-duplex

D. Broadcast only

Correct Answer: 1

Explanation: Full duplex allows communication in both directions at the same time.

Q7. (Medium) Satellite latency nature?

A. High delay

B. Zero delay

C. Negative delay

D. No delay

Correct Answer: 1

Explanation: Satellite communication involves long distances, causing high delay.

Q8. (Medium) Circuit switching used in?

A. Telephone

B. Internet core

C. WiFi

D. Bluetooth

Correct Answer: 1

Explanation: Traditional telephone systems use circuit switching.

Q9. (Medium) Packet switching is?

A. Dedicated

B. Connectionless

C. Static

D. Reserved

Correct Answer: 2

Explanation: Packet switching is connectionless where data is sent in packets independently.

Q10. (Medium) Routing determines?

A. Best path

B. Signal strength

C. Frame size

D. Encryption

Correct Answer: 1

Explanation: Routing determines the best path for data transmission.

Q11. (Medium) Packet contains?

A. Only footer

B. Only header

C. Only data

D. Header & data

Correct Answer: 4

Explanation: A packet consists of both header (control info) and data (payload).

Q12. (Medium) Store-and-forward used in?

A. Coax only

B. Circuit switching

C. Analog lines

D. Packet switching

Correct Answer: 4

Explanation: Store-and-forward technique is used in packet switching.

Q13. (Medium) Routing table stores?

A. Signals

B. Passwords

C. Frames

D. Paths

Correct Answer: 4

Explanation: Routing tables store paths/routes to destinations.

Q14. (Medium) Packet switching used in?

A. PSTN

B. Internet

C. Radio

D. TV

Correct Answer: 2

Explanation: The Internet uses packet switching.

Q15. (Medium) Control signaling does?

A. Compresses data

B. Sends payload

C. Manages connection

D. Encrypts

Correct Answer: 3

Explanation: Control signaling is used to manage and establish connections.

Q16. (Medium) Shortest path algorithm finds?

A. Random path

B. Longest path

C. Minimum cost path

D. Fixed path

Correct Answer: 3

Explanation: Shortest path algorithms find minimum cost paths.

Q17. (Medium) Parity bit used for?

A. Encryption

B. Error detection

C. Routing

D. Compression

Correct Answer: 2

Explanation: Parity bits are used for detecting errors in transmission.

Q18. (Medium) Checksum purpose?

A. Switching

B. Addressing

C. Error detection

D. Timing

Correct Answer: 3

Explanation: Checksum is used for error detection in data communication.

Q19. (Medium) CRC uses?

A. Polynomial division

B. Addition

C. Multiplication

D. Subtraction

Correct Answer: 1

Explanation: CRC uses polynomial division for error detection.

Q20. (Medium) Hamming code corrects?

A. Single-bit error

B. Two-bit error

C. Burst only

D. No error

Correct Answer: 1

Explanation: Hamming code can detect and correct single-bit errors.

Q21. (Medium) Even parity means?

A. No 1s

B. Odd number of 1s

C. Even number of 1s

D. All 0s

Correct Answer: 3

Explanation: Even parity ensures total number of 1s is even.

Q22. (Medium) CRC detects?

A. Only 1-bit

B. Burst errors

C. Timing errors

D. Routing loops

Correct Answer: 2

Explanation: CRC is effective in detecting burst errors.

Q23. (Medium) Checksum layer?

A. Transport layer

B. Physical

C. Session

D. Presentation

Correct Answer: 1

Explanation: Checksum is used at the transport layer (e.g., TCP/UDP).

Q24. (Medium) Hamming adds?

A. Trailer bits

B. Payload bits

C. Header bits

D. Redundant bits

Correct Answer: 4

Explanation: Hamming code adds redundant bits for error correction.

Q25. (Medium) Stop & Wait sends?

A. Multiple frames

B. One frame

C. No frames

D. Infinite frames

Correct Answer: 2

Explanation: Stop-and-Wait protocol sends one frame at a time.

Q26. (Medium) Go-Back-N retransmits?

A. From error onward

B. Only error frame

C. All frames always

D. No frames

Correct Answer: 1

Explanation: Go-Back-N retransmits from the error frame onward.

Q27. (Medium) Window size Stop & Wait?

A. 8

B. 2

C. 4

D. 1

Correct Answer: 4

Explanation: Stop-and-Wait has window size of 1.

Q28. (Medium) ACK stands for?

A. Access key

B. Acknowledgment

C. Active kernel

D. Auto keep

Correct Answer: 2

Explanation: ACK stands for acknowledgment.

Q29. (Medium) Flow control prevents?

A. Encryption

B. Routing loops

C. Buffer overflow

D. Address clash

Correct Answer: 3

Explanation: Flow control prevents buffer overflow at receiver.

Q30. (Medium) GBN uses?

A. Cumulative ACK

B. Selective ACK

C. No ACK

D. Negative ACK only

Correct Answer: 1

Explanation: GBN uses cumulative acknowledgments.

Q31. (Medium) Timeout causes?

A. Routing

B. Shutdown

C. Retransmission

D. Fragmentation

Correct Answer: 3

Explanation: Timeout leads to retransmission of frames.

Q32. (Medium) Sliding window used in?

A. Switching

B. Routing

C. Flow control

D. Encoding

Correct Answer: 3

Explanation: Sliding window protocol is used for flow control.

Q33. (Medium) ALOHA is?

A. Error code

B. Routing protocol

C. Random access protocol

D. Encryption

Correct Answer: 3

Explanation: ALOHA is a random access protocol.

Q34. (Medium) CSMA/CD used in?

A. WiFi

B. Ethernet

C. Satellite

D. Bluetooth

Correct Answer: 2

Explanation: CSMA/CD is used in Ethernet networks.

Q35. (Medium) CSMA/CA used in?

A. Ethernet

B. WiFi

C. Fiber

D. DSL

Correct Answer: 2

Explanation: CSMA/CA is used in WiFi networks.

Q36. (Medium) IPv4 size?

A. 16 bits

B. 64 bits

C. 128 bits

D. 32 bits

Correct Answer: 4

Explanation: IPv4 uses 32-bit addressing.

Q37. (Medium) IPv6 size?

A. 64 bits

B. 32 bits

C. 128 bits

D. 48 bits

Correct Answer: 3

Explanation: IPv6 uses 128-bit addressing.

Q38. (Medium) ARP maps?

A. MAC to port

B. MAC to IP

C. IP to port

D. IP to MAC

Correct Answer: 4

Explanation: ARP maps IP address to MAC address.

Q39. (Medium) DHCP assigns?

A. IP address

B. MAC

C. Port

D. Frame

Correct Answer: 1

Explanation: DHCP automatically assigns IP addresses.

Q40. (Medium) NAT does?

A. Encrypts data

B. Public to MAC

C. Private to public IP

D. Routes packets

Correct Answer: 3

Explanation: NAT converts private IP addresses to public IP addresses.

Class Test 2 Question with Answers

Q1. A communication satellite mainly acts as a:

A. Signal generator

B. Signal amplifier and repeater

C. Data storage device

D. Router

Correct Answer: Signal amplifier and repeater

Explanation: Signal amplifier and repeater is correct because it directly satisfies the concept or definition asked in the question.

Q2. Which orbit is commonly used for TV broadcasting satellites?

A. LEO (Low Earth Orbit)

B. MEO (Medium Earth Orbit)

C. GEO (Geostationary Orbit)

D. Polar Orbit

Correct Answer: GEO (Geostationary Orbit)

Explanation: GEO (Geostationary Orbit) is correct because it directly satisfies the concept or definition asked in the question.

Q3. In circuit switching, a dedicated path is established:

A. After data transmission

B. Before data transmission

C. During data transmission

D. Only when error occurs

Correct Answer: Before data transmission

Explanation: Before data transmission is correct because it directly satisfies the concept or definition asked in the question.

Q4. In circuit switching, once the path is established:

A. Data is sent in packets

B. Bandwidth is shared

C. Communication occurs continuously

D. Connection is temporary per packet

Correct Answer: Communication occurs continuously

Explanation: Communication occurs continuously is correct because it directly satisfies the concept or definition asked in the question.

Q5. In circuit switching, "Alternate Routing" is primarily used to:

A. Increase data rate

B. Reduce hardware cost

C. Handle traffic overflows

D. Encrypt data

Correct Answer: Handle traffic overflows

Explanation: Handle traffic overflows is correct because it directly satisfies the concept or definition asked in the question.

Q6. In packet switching, the principle of "Store and Forward" means:

A. Data is stored until the user deletes it

B. A switch receives the entire packet before forwarding

C. Packets are stored on the user's PC

D. Data is only sent when the network is empty

Correct Answer: A switch receives the entire packet before forwarding

Explanation: A switch receives the entire packet before forwarding is correct because it directly satisfies the concept or definition asked in the question.

Q7. In a Virtual Circuit network, the identifier used to route packets is called:

A. IP Address

B. MAC Address

C. VCI (Virtual Circuit Identifier)

D. Port Number

Correct Answer: VCI (Virtual Circuit Identifier)

Explanation: VCI (Virtual Circuit Identifier) is correct because it directly satisfies the concept or definition asked in the question.

Q8. Which of the following can detect all single-bit errors but fails to detect even numbers of bit errors?

A. Checksum

B. Simple Parity Check

C. CRC

D. Hamming Code

Correct Answer: Simple Parity Check

Explanation: Simple Parity Check is correct because it directly satisfies the concept or definition asked in the question.

Q9. In Checksum, if the sum of the data segments is 1010, what will be the checksum value sent by the sender?

A. 1010

B. 101

C. 1111

D. 0

Correct Answer: 101

Explanation: 101 is correct because it directly satisfies the concept or definition asked in the question.

Q10. The Cyclic Redundancy Check (CRC) is based on which mathematical operation?

A. Polynomial multiplication

B. Binary (Modulo-2) division

C. One's complement subtraction

D. Matrix inversion

Correct Answer: Binary (Modulo-2) division

Explanation: Binary (Modulo-2) division is correct because it directly satisfies the concept or definition asked in the question.

Q11. What is the Hamming distance between the codewords 10101 and 11100?

A. 1

B. 2

C. 3

D. 5

Correct Answer: 2

Explanation: 2 is correct because it directly satisfies the concept or definition asked in the question.

Q12. Minimum sequence numbers required for Stop-and-Wait?

A. 1

B. 2

C. Unlimited

D. Depends on window

Correct Answer: 2

Explanation: 2 is correct because it directly satisfies the concept or definition asked in the question.

Q13. What does the sender do in Go-Back-N if a timer expires?

A. Retransmit one

B. Wait for request

C. Retransmit lost + subsequent

D. Drop connection

Correct Answer: Retransmit lost + subsequent

Explanation: Retransmit lost + subsequent is correct because it directly satisfies the concept or definition asked in the question.

Q14. Attaching an ACK to an outgoing data frame is:

A. Pipelining

B. Encapsulation

C. Piggybacking

D. Buffering

Correct Answer: Piggybacking

Explanation: Piggybacking is correct because it directly satisfies the concept or definition asked in the question.

Q15. Primary advantage of Selective Repeat over GBN?

A. Simpler

B. Smaller buffer

C. Only damaged frame re-sent

D. No sequence #

Correct Answer: Only damaged frame re-sent

Explanation: Only damaged frame re-sent is correct because it directly satisfies the concept or definition asked in the question.

Q16. What to do in CSMA/CD after detecting collision?

A. Stop + jam signal

B. Continue

C. Restart

D. Wait 1s

Correct Answer: Stop + jam signal

Explanation: Stop + jam signal is correct because it directly satisfies the concept or definition asked in the question.

Q17. Period waited in CSMA/CA after medium is idle?

A. Backoff time

B. IFS

C. Collision window

D. RTT

Correct Answer: IFS

Explanation: IFS is correct because it directly satisfies the concept or definition asked in the question.

Q18. What is the total size of an IPv4 address?

A. 32 bits

B. 64 bits

C. 128 bits

D. 48 bits

Correct Answer: 32 bits

Explanation: 32 bits is correct because it directly satisfies the concept or definition asked in the question.

Q19. IPv6 header is simplified by removing which field?

A. Source Addr

B. Dest Addr

C. Header Checksum

D. Version

Correct Answer: Header Checksum

Explanation: Header Checksum is correct because it directly satisfies the concept or definition asked in the question.

Q20. Primary purpose of the TTL field in IPv4?

A. Measure speed

B. Prevent infinite loops

C. Prioritize voice

D. Sync clocks

Correct Answer: Prevent infinite loops

Explanation: Prevent infinite loops is correct because it directly satisfies the concept or definition asked in the question.

Q21. Which is TRUE regarding IPv4/IPv6 comparison?

A. IPv6 header is smaller

B. IPv6 is 4x larger

C. IPv6 has 40-byte fixed header

D. IPv4 no fragmentation

Correct Answer: IPv6 has 40-byte fixed header

Explanation: IPv6 has 40-byte fixed header is correct because it directly satisfies the concept or definition asked in the question.

Q22. DHCP improvement over BOOTP?

A. Perm. MAC

B. Fixed Names

C. Leased IP

D. Encryption

Correct Answer: Leased IP

Explanation: Leased IP is correct because it directly satisfies the concept or definition asked in the question.

Q23. In "Direct Delivery", source and destination are:

A. Different net

B. Same net

C. Multi-router

D. Diff protocol

Correct Answer: Same net

Explanation: Same net is correct because it directly satisfies the concept or definition asked in the question.

Q24. NAT internal networks typically use:

A. Global unicast

B. Public

C. Private

D. Anycast

Correct Answer: Private

Explanation: Private is correct because it directly satisfies the concept or definition asked in the question.

Q25. The "Count-to-Infinity" problem is associated with which protocol?

A. Link State

B. Distance Vector

C. Shortest Path

D. Flooding

Correct Answer: Distance Vector

Explanation: Distance Vector is correct because it directly satisfies the concept or definition asked in the question.

Q26. Which is an "Open Loop" congestion control policy?

A. Backpressure

B. Choke packets

C. Retransmission policy

D. Implicit signaling

Correct Answer: Retransmission policy

Explanation: Retransmission policy is correct because it directly satisfies the concept or definition asked in the question.

Q27. In Link State Routing, nodes distribute info via:

A. Distance Vector

B. Link State Packet (LSP)

C. Hello Packet

D. Routing Table

Correct Answer: Link State Packet (LSP)

Explanation: Link State Packet (LSP) is correct because it directly satisfies the concept or definition asked in the question.

Q28. Which layer provides process-to-process communication?

A. Network

B. Data Link

C. Transport

D. Application

Correct Answer: Transport

Explanation: Transport is correct because it directly satisfies the concept or definition asked in the question.

Q29. Gathering data from different apps to the network layer is:

A. Demultiplexing

B. Segmentation

C. Multiplexing

D. Encapsulation

Correct Answer: Multiplexing

Explanation: Multiplexing is correct because it directly satisfies the concept or definition asked in the question.

Q30. Primary role of "Buffering" in transport layer?

A. Speed up cables

B. Store if app not ready

C. Encrypt data

D. Checksum

Correct Answer: Store if app not ready

Explanation: Store if app not ready is correct because it directly satisfies the concept or definition asked in the question.

Q31. Ports 0–1023 are known as:

A. Registered

B. Dynamic

C. Well-known

D. Private

Correct Answer: Well-known

Explanation: Well-known is correct because it directly satisfies the concept or definition asked in the question.

Q32. In FTP Passive Mode, who initiates the data connection?

A. FTP Server

B. FTP Client

C. Firewall

D. Router

Correct Answer: Firewall

Explanation: Firewall is correct because it directly satisfies the concept or definition asked in the question.

Q33. Protocol used with SIP for real-time media streams?

A. TCP

B. RTP

C. FTP

D. ICMP

Correct Answer: FTP

Explanation: FTP is correct because it directly satisfies the concept or definition asked in the question.

Q34. Port used for modern SMTP email submission?

A. 25

B. 110

C. 587

D. 143

Correct Answer: 587

Explanation: 587 is correct because it directly satisfies the concept or definition asked in the question.

Q35. The process used by TCP to establish a connection is known as:

A. Flow Control

B. Error Control

C. Three-way handshake

D. Multiplexing

Correct Answer: Three-way handshake

Explanation: Three-way handshake is correct because it directly satisfies the concept or definition asked in the question.

Q36. Which is NOT typically considered a parameter for QoS?

A. Reliability

B. Delay

C. Jitter

D. IP Version

Correct Answer: IP Version

Explanation: IP Version is correct because it directly satisfies the concept or definition asked in the question.

Q37. Why is UDP preferred for real-time video conferencing?

A. More secure

B. Lower overhead / no retrans. delay

C. Guaranteed delivery

D. Fiber only

Correct Answer: Lower overhead / no retrans. delay

Explanation: Lower overhead / no retrans. delay is correct because it directly satisfies the concept or definition asked in the question.

Q38. Geostationary satellite completes orbit in?

A. 24 hours

B. 12 hours

C. 48 hours

D. 6 hours

Correct Answer: 24 hours

Explanation: 24 hours is correct because it directly satisfies the concept or definition asked in the question.

Q39. Routing decides?

A. Best path

B. Frame size

C. Speed

D. Error

Correct Answer: Best path

Explanation: Best path is correct because it directly satisfies the concept or definition asked in the question.

Q40. Parity bit is used for?

A. Compression

B. Routing

C. Encryption

D. Error detection

Correct Answer: Error detection

Explanation: Error detection is correct because it directly satisfies the concept or definition asked in the question.

Q41. Hamming code corrects?

A. No error

B. Double-bit error

C. Burst error

D. Single-bit error

Correct Answer: Single-bit error

Explanation: Single-bit error is correct because it directly satisfies the concept or definition asked in the question.

Q42. Go-Back-N retransmits?

A. Only one frame

B. From error onward

C. All frames always

D. No retransmission

Correct Answer: From error onward

Explanation: From error onward is correct because it directly satisfies the concept or definition asked in the question.

Q43. ALOHA is?

A. Routing protocol

B. Random access protocol

C. Error control

D. Switching

Correct Answer: Random access protocol

Explanation: Random access protocol is correct because it directly satisfies the concept or definition asked in the question.

Q44. IPv4 size?

A. 16 bits

B. 64 bits

C. 128 bits

D. 32 bits

Correct Answer: 32 bits

Explanation: 32 bits is correct because it directly satisfies the concept or definition asked in the question.

Q45. ARP maps?

A. IP to MAC

B. MAC to IP

C. IP to Port

D. MAC to Port

Correct Answer: IP to MAC

Explanation: IP to MAC is correct because it directly satisfies the concept or definition asked in the question.

Q46. NAT translates?

A. Encrypts data

B. Public to MAC

C. Private to public IP

D. Routes packets

Correct Answer: Private to public IP

Explanation: Private to public IP is correct because it directly satisfies the concept or definition asked in the question.

Q47. Link state routing uses?

A. Partial info

B. Complete topology

C. None

D. Random

Correct Answer: Complete topology

Explanation: Complete topology is correct because it directly satisfies the concept or definition asked in the question.

Computer network sample question Bank.

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- A. Signal generator
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- C. Jitter
- D. IP Version

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- A. More secure
- B. Lower overhead / no retrans. delay
- C. Guaranteed delivery
- D. Fiber only

Q38. The device in a satellite that receives, amplifies, and retransmits signals is called:

- A. Antenna
- B. Modulator
- C. Transponder
- D. Oscillator

Q39. The time delay in satellite communication is mainly due to:

- A. Noise
- B. Long distance between earth and satellite
- C. Signal distortion
- D. Frequency mismatch

Q40. Which of the following is an example of circuit switching?

- A. Internet
- B. Email
- C. Traditional telephone network
- D. FTP

Q41. Which phase is NOT part of circuit switching?

- A. Setup phase
- B. Data transfer phase
- C. Termination phase
- D. Routing phase

Q42. Which type of signaling involves sending control signals over the same channel as the user data?

- A. Out-of-band signaling
- B. Common Channel Signaling
- C. In-band signaling
- D. Packet signaling

Q43. Which switching technique treats each packet as an independent entity with no prior path setup?

- A. Circuit switching
- B. Virtual circuit switching
- C. Datagram switching
- D. Message switching

Q44. What happens to a packet-switched network when the traffic load exceeds the network capacity?

- A. The connection is dropped
- B. Congestion and increased delay occur
- C. The network speeds up
- D. Data is automatically deleted

Q45. In even parity, the parity bit is set such that:

- A. Total number of 0s is even
- B. Total number of 1s is even
- C. Total number of bits is odd
- D. Number of 1s is always odd

Q46. If the divisor (generator polynomial) for CRC is x3+x+1, what is its binary representation?

- A. 1011
- B. 1101
- C. 111
- D. 1001

Q47. To provide error correction for a 7-bit data unit using Hamming code, what is the minimum number of redundant bits (r) required?

- A. 3
- B. 4
- C. 5
- D. 7

Q48. What is the primary purpose of flow control?

- A. Detect bit errors
- B. Prevent overwhelming receiver
- C. Shortest path
- D. Encrypt data

Q49. Which statement is true for Go-Back-N ARQ?

- A. Receiver accepts out-of-order frames
- B. Sender retransmits only lost frame
- C. Sender retransmits all frames after an error
- D. Window size is 2m−12^{m-1}2m−1

Q50. The mechanism involving retransmission of data is:

- A. Framing
- B. ARQ
- C. CSMA/CD
- D. Piggybacking

Q51. Efficiency formula for sliding window (Tp=Prop, Tt=Trans)?

- A. 1/(1+Tp/Tt)
- B. W/(1+2(Tp/Tt))
- C. Tt/(Tt+Tp)
- D. 1/(1+2(Tp/Tt))

Q52. what is Pure ALOHA vulnerable time

- A. 2 X Tfr
- B. 1 X Tfr
- C. 2 X T/fr
- D. 2 / Tfr

Q53. CSMA/CD is primarily used in:

- A. Wireless LAN

- B. Wired Ethernet
- C. Satellite
- D. Fiber

Q54. Probability of transmission in 1-persistent CSMA if idle?

- A. 0
- B. 0.5
- C. 1
- D. 0.1

Q55. What is the total size of an IPv6 address?

- A. 32 bits
- B. 64 bits
- C. 128 bits
- D. 256 bits

Q56. Field in IPv4 used to reassemble fragments?

- A. TTL
- B. Identification
- C. Type of Service
- D. Protocol

Q57. Extra functions like security in IPv6 use:

- A. Options field
- B. Extension Headers
- C. Payload length
- D. ICMPv4

Q58. ARP is used to find the _______ when the _______ is known.

- A. IP; MAC
- B. MAC; IP
- C. Name; IP
- D. Port; MAC

Q59. Transport protocol and ports used by DHCP?

- A. TCP; 80/443
- B. UDP; 67/68
- C. TCP; 20/21
- D. UDP; 53/54

Q60. BOOTP is considered a _______ protocol.

- A. Dynamic
- B. Static
- C. Connection
- D. Multicast

Q61. Indirect delivery always involves at least one:

- A. Switch
- B. Router
- C. Repeater
- D. Bridge

Q62. In Flooding, what happens when a packet arrives at a node?

- A. Shortest path only
- B. Sent to all except arrival line
- C. Stored
- D. Discarded

Q63. The Leaky Bucket algorithm produces a _______ output rate.

- A. Variable
- B. Constant
- C. Bursty
- D. Random

Q64. Controlling the rate of packets sent into the network is called:

- A. Routing
- B. Traffic Shaping
- C. Switching
- D. Multiplexing

Q65. Address used to identify a process on a host?

- A. IP Address
- B. MAC Address
- C. Port Number
- D. Logical Address

Q66. Delivering segments to the correct socket at destination?

- A. Multiplexing
- B. Demultiplexing
- C. Routing
- D. Flow Control

Q67. Typical range for "Ephemeral" (temporary) ports?

- A. 0–1023
- B. 1024–49151
- C. 49152–65535
- D. 80–443

Q68. HTTP method used to request data without changes?

- A. POST
- B. PUT
- C. GET
- D. DELETE

Q69. Protocol preferred for multi-device email sync?

- A. POP3
- B. SMTP
- C. IMAP

- D. FTP

Q70. "404 Not Found" belongs to which HTTP category?

- A. Informational
- B. Success
- C. Client Error
- D. Server Error

Q71. Which of the following is a key characteristic of the TCP protocol?

- A. Connectionless
- B. Unreliable delivery
- C. Connection-oriented and reliable
- D. No acknowledgments

Q72. Which SCTP feature supports multiple IP addresses at each end?

- A. Multi-streaming
- B. Multi-homing
- C. Full-duplex
- D. Four-way handshake

Q73. The IntServ model for QoS relies on which protocol to reserve resources?

- A. ICMP
- B. RSVP
- C. DHCP
- D. DNS

Q74. Which QoS scheduling technique ensures fair bandwidth distribution among all active flows?

- A. FIFO
- B. Priority Queuing
- C. Fair Queuing
- D. Round Robin

Q75. Which orbit is commonly used for TV broadcasting satellites?

- A. LEO (Low Earth Orbit)
- B. MEO (Medium Earth Orbit)
- C. GEO (Geostationary Orbit)
- D. Polar Orbit

Q76. A switched network is defined as a network:

- A. With fixed paths only
- B. That uses multiple possible paths for data transmission
- C. Without routers
- D. That uses only wireless communication

Q77. One major disadvantage of circuit switching is:

- A. High delay in setup
- B. Inefficient use of bandwidth
- C. Packet loss
- D. No dedicated path

Q78. Which routing strategy in circuit-switched networks uses a single, permanent route for each source-destination pair?

- A. Dynamic routing
- B. Fixed routing
- C. Flooding
- D. Adaptive routing

Q79. What is the primary purpose of control signaling in a circuit-switched network?

- A. To transmit user data
- B. To establish, manage, and terminate connections
- C. To encrypt data
- D. To compress data

Q80. What is the primary advantage of Packet Switching over Circuit Switching?

- A. Guaranteed bandwidth
- B. Lower latency for voice
- C. More efficient use of link capacity
- D. Simpler routing tables

Q81. What happens during the teardown phase in circuit switching?

- A. Data is transmitted
- B. Resources are reserved
- C. Connection is established
- D. Resources are released

Q82. In the Checksum error detection method, the sender adds the data units using which type of arithmetic?

- A. Binary addition
- B. Two's complement
- C. One's complement
- D. Modulo-2

Q83. In CRC, if the generator polynomial is of degree n, how many zero bits are appended to the data unit before starting the division?

- A. n+1
- B. n−1
- C. n
- D. 2n

Q84. In a Hamming code, at which positions are the redundant (parity) bits placed?

- A. At the beginning
- B. At the end
- C. At positions that are powers of 2 (1, 2, 4, 8...)
- D. Randomly

Q85. What is the main disadvantage of Stop-and-Wait ARQ?

- A. Too complex

- B. High CPU power
- C. Inefficient for high propagation delay
- D. Cannot detect lost frames

Q86. Size of the receiver window in Go-Back-N ARQ?

- A. 1
- B. Same as sender
- C. 2m
- D. Half of sender

Q87. Which uses "Cumulative Acknowledgment"?

- A. Stop-and-Wait
- B. Sliding Window (GBN)
- C. UDP
- D. Simple Protocol

Q88. The maximum sender window size in Selective Repeat ARQ is:

- A. Equal to total sequence numbers
- B. Half of total sequence numbers
- C. One less than total sequence numbers
- D. Twice the sequence numbers

Q89. Improvement in Slotted ALOHA over Pure ALOHA?

- A. Collision detect
- B. Doubles throughput (36.8%)
- C. No collisions
- D. No sync

Q90. Why use CSMA/CA in wireless instead of CD?

- A. Too expensive
- B. Cannot reliably detect CD
- C. Signal speed
- D. CA is slower

Q91. Vulnerable time for Pure ALOHA?

- A. T
- B. 2 X Tfr
- C. T/2
- D. Prop. delay

Q92. Minimum header length of an IPv4 datagram?

- A. 10 bytes
- B. 20 bytes
- C. 40 bytes
- D. 60 bytes

Q93. Notation typically used for IPv6 addresses?

- A. Dotted-decimal
- B. Binary
- C. Hexadecimal-colon
- D. Octal

Q94. Reserved IPv4 address for loopback testing?

- A. 192.168.1.1
- B. 127.0.0.1
- C. 0.0.0.0
- D. 255.255.255.255

Q95. Protocol for diskless workstations to find their IP?

- A. ARP
- B. ICMP
- C. RARP
- D. IGMP

Q96. Primary use of NAT?

- A. Global routing
- B. Conserve IPv4
- C. Speed
- D. Replace L2

Q97. ARP Request is _______; ARP Reply is _______.

- A. Unicast; Bcast
- B. Bcast; Unicast
- C. Mcast; Unicast
- D. Bcast; Mcast

Q98. Which routing algorithm uses Dijkstra's algorithm for the shortest path?

- A. Distance Vector
- B. Link State
- C. Flooding
- D. Static

Q99. Congestion Control is concerned with:

- A. Single sender/receiver
- B. Entire network traffic/capacity
- C. Error detection
- D. IP assignment

Q100. Advantage of Token Bucket over Leaky Bucket?

- A. Easier implementation
- B. Allows bursty data
- C. Discards excess
- D. No buffer

Q101. Distance Vector routing is based on which algorithm?

- A. Bellman-Ford
- B. Dijkstra
- C. Greedy

- D. Prim's

Q102. A "Socket" is a combination of:

- A. MAC & IP
- B. IP & Port
- C. Port & Protocol
- D. Name & IP

Q103. Flow control in the Transport Layer is performed:

- A. Node-to-node
- B. End-to-end
- C. Hop-to-hop
- D. Physical link

Q104. Mechanism to adjust data amount a sender can transmit?

- A. Token Bucket
- B. Sliding Window
- C. Stop & Wait
- D. Binary Backoff

Q105. DNS record used to map a domain name to IPv4?

- A. AAAA
- B. MX
- C. A record
- D. CNAME

Q106. Primary role of SIP in VoIP?

- A. Carry voice data
- B. Encode signals
- C. Manage sessions
- D. Physical link

Q107. Variation in the delay of received VoIP packets?

- A. Latency
- B. Jitter
- C. Bandwidth
- D. Attenuation

Q108. What is the fixed size of a UDP header?

- A. 8 bytes
- B. 20 bytes
- C. 40 bytes
- D. 60 bytes

Q109. Unlike TCP, SCTP uses a ________ handshake to establish associations.

- A. Two-way
- B. Three-way
- C. Four-way
- D. Five-way

Q110. In DiffServ, packets are marked with a priority value called:

- A. MAC address
- B. Port number
- C. DSCP
- D. Sequence number

Q111. In Pure ALOHA, what is the time interval in which transmitted frames are susceptible to collision?

A. 2 X Tfr
B. 1 X Tfr
C. 2 X T/fr
D. 2 / Tfr

Q112

Q1. Circuit switching requires?

1. Dynamic routing

2. Packet path

3. Dedicated path

4. Broadcasting

Q2. Routing decides?

1. Best path

2. Frame size

3. Speed

4. Error

Q3. Checksum verifies?

1. Addressing

2. Routing

3. Data integrity

4. Timing

Q4. CSMA/CD is used in?

1. Ethernet

2. WiFi

3. Bluetooth

4. Satellite

Q5. IPv6 size?

1. 64 bits

2. 32 bits

3. 128 bits

4. 48 bits

Q6. IPv4 header is?

1. Static

2. Fixed

3. Random

4. Variable

Q7. RARP maps?

2. IP to MAC

3. Port to IP

4. MAC to IP

Q8. Shortest path algorithm finds?

1. Fixed path

2. Longest path

3. Random path

4. Minimum cost path

Q9. Flooding sends packets to?

1. One node

2. All nodes

3. Random node

Q10. Leaky bucket controls?

1. Traffic rate

2. Routing

3. Encryption

4. Error

Q11. Packet switching is?

1. Reserved

2. Dedicated

3. Connectionless

4. Fixed

Q12. Control signaling is used for?

1. Data transfer

2. Connection management

3. Encryption

4. Compression

Q13. CRC is based on?

1. Polynomial division

2. Addition

3. Subtraction

4. Multiplication

Q14. Stop and Wait ARQ sends?

1. Random frames

2. Multiple frames

3. No frames

4. One frame at a time

Q15. Selective Repeat retransmits?

1. All frames

2. Only error frames

3. No frames

4. Duplicate frames

Q16. CSMA/CA is used in?

1. WiFi

2. Ethernet

3. Fiber

4. DSL

Q17. IPv6 header is?

1. Dynamic

2. Variable

3. Random

4. Fixed

Q18. DHCP provides?

1. Frame

2. MAC address

3. Port

4. IP address

Q19. Distance vector routing uses?

1. Bandwidth

2. Hop count

3. Delay

4. Error

Q20. Token bucket allows?

1. No traffic

2. Burst traffic

3. Constant only

4. Random

Q21 Geostationary satellite completes orbit in?

1. 24 hours
2. 12 hours
3. 48 hours
4. 6 hours

Computer Networks - Set 2

Section 1: 1 Mark Questions

1. Which uses "Cumulative Acknowledgment"?

A) Stop-and-Wait

B) Sliding Window (GBN)

C) UDP

D) Simple Protocol

2. Minimum sequence numbers required for Stop-and-Wait?

A) 1

B) 2

C) Unlimited

D) Depends on window

3. Address used to identify a process on a host?

A) IP Address

B) MAC Address

C) Port Number

D) Logical Address

4. Congestion Control is concerned with:

A) Single sender/receiver

B) Entire network traffic/capacity

C) Error detection

D) IP assignment

5. In a Hamming code, at which positions are the redundant (parity) bits placed?

A) At the beginning

B) At the end

C) At positions that are powers of 2 (1, 2, 4, 8...)

D) Randomly

6. DHCP improvement over BOOTP?

A) Permanent MAC Address.

B) Fixed Names

C) Leased IP

D) Encryption

7. Which is TRUE regarding IPv4/IPv6 comparison?

A) IPv6 header is smaller

B) IPv6 is 4x larger

C) IPv6 has 40-byte fixed header

D) IPv4 no fragmentation

8. The device in a satellite that receives, amplifies, and retransmits signals is called:

A) Antenna

B) Modulator

C) Transponder

D) Oscillator

9. The process used by TCP to establish a connection is known as:

A) Flow Control

B) Error Control

C) Three-way handshake

D) Multiplexing

10. Ports 0–1023 are known as:

A) Registered

B) Dynamic

C) Well-known

D) Private

11. "404 Not Found" belongs to which HTTP category?

A) Informational

B) Success

C) Client Error

D) Server Error

12. Which is an "Open Loop" congestion control policy?

A) Backpressure

B) Choke packets

C) Retransmission policy

D) Implicit signaling

13. The maximum sender window size in Selective Repeat ARQ is:

A) Equal to total sequence numbers

B) Half of total sequence numbers

C) One less than total sequence numbers

D) Twice the sequence numbers

14. What is the primary purpose of control signaling in a circuit-switched network?

A) To transmit user data

B) To establish, manage, and terminate connections

C) To encrypt data

D) To compress data

15. Efficiency formula for sliding window (Tp=Prop, Tt=Trans)?

A) 1/(1+Tp/Tt)

B) W/(1+2(Tp/Tt))

C) Tt/(Tt+Tp)

D) 1/(1+2(Tp/Tt))

16. In circuit switching, once the path is established:

A) Data is sent in packets

B) Bandwidth is shared

C) Communication occurs continuously

D) Connection is temporary per packet

17. Which switching technique treats each packet as an independent entity with no prior path setup?

A) Circuit switching

B) Virtual circuit switching

C) Datagram switching

D) Message switching

18. In circuit switching, a dedicated path is established:

A) After data transmission

B) Before data transmission

C) During data transmission

D) Only when error occurs

19. A communication satellite mainly acts as a:

A) Signal generator

B) Signal amplifier and repeater

C) Data storage device

D) Router

20. What is the total size of an IPv4 address?

A) 32 bits

B) 64 bits

C) 128 bits

D) 48 bits

Section 2: 2 Mark Questions

1. Parity bit is used for?

A) Compression

B) Routing

C) Encryption

D) Error detection

2. Shortest path algorithm finds?

A) Fixed path

B) Longest path

C) Random path

D) Minimum cost path

3. Routing decides?

A) Best path

B) Frame size

C) Speed

D) Error

4. Circuit switching requires?

A) Dynamic routing

B) Packet path

C) Dedicated path

D) Broadcasting

5. IPv6 header is?

A) Dynamic

B) Variable

C) Random

D) Fixed

6. Stop and Wait ARQ sends?

A) Random frames

B) Multiple frames

C) No frames

D) One frame at a time

7. IPv4 size?

A) 16 bits

B) 64 bits

C) 128 bits

D) 32 bits

8. Routing decides?

A) Best path

B) Frame size

C) Speed

D) Error

9. Leaky bucket controls?

A) Traffic rate

B) Routing

C) Encryption

D) Error

10. CSMA/CA is used in?

A) WiFi

B) Ethernet

C) Fiber

D) DSL

Computer Networks - Set 1

Section 1: 1 Mark Questions

1. Minimum header length of an IPv4 datagram?

A) 10 bytes

B) 20 bytes

C) 40 bytes

D) 60 bytes

2. What is the total size of an IPv4 address?

A) 32 bits

B) 64 bits

C) 128 bits

D) 48 bits

3. NAT internal networks typically use:

A) Global unicast

B) Public

C) Private

D) Anycast

4. Which of the following is an example of circuit switching?

A) Internet

B) Email

C) Traditional telephone network

D) FTP

5. Which is NOT typically considered a parameter for QoS?

A) Reliability

B) Delay

C) Jitter

D) IP Version

6. If the divisor (generator polynomial) for CRC is x3+x+1, what is its binary representation?

A) 1011

B) 1101

C) 111

D) 1001

7. In Link State Routing, nodes distribute info via:

A) Distance Vector

B) Link State Packet (LSP)

C) Hello Packet

D) Routing Table

8. Controlling the rate of packets sent into the network is called:

A) Routing

B) Traffic Shaping

C) Switching

D) Multiplexing

9. Attaching an ACK to an outgoing data frame is:

A) Pipelining

B) Encapsulation

C) Piggybacking

D) Buffering

10. In packet switching, the principle of "Store and Forward" means:

A) Data is stored until the user deletes it

B) A switch receives the entire packet before forwarding

C) Packets are stored on the user's PC

D) Data is only sent when the network is empty

11. Probability of transmission in 1-persistent CSMA if idle?

A) 0

B) 0.5

C) 1

D) 0.1

12. In Flooding, what happens when a packet arrives at a node?

A) Shortest path only

B) Sent to all except arrival line

C) Stored

D) Discarded

13. Unlike TCP, SCTP uses a _________ handshake to establish associations.

A) Two-way

B) Three-way

C) Four-way

D) Five-way

14. Minimum sequence numbers required for Stop-and-Wait?

A) 1

B) 2

C) Unlimited

D) Depends on window

15. The Cyclic Redundancy Check (CRC) is based on which mathematical operation?

A) Polynomial multiplication

B) Binary (Modulo-2) division

C) One's complement subtraction

D) Matrix inversion

16. In a Virtual Circuit network, the identifier used to route packets is called:

A) IP Address

B) MAC Address

C) VCI (Virtual Circuit Identifier)

D) Port Number

17. CSMA/CD is primarily used in:

A) Wireless LAN

B) Wired Ethernet

C) Satellite

D) Fiber

18. Which routing strategy in circuit-switched networks uses a single, permanent route for each source-destination pair?

A) Dynamic routing

B) Fixed routing

C) Flooding

D) Adaptive routing

19. In even parity, the parity bit is set such that:

A) Total number of 0s is even

B) Total number of 1s is even

C) Total number of bits is odd

D) Number of 1s is always odd

20. Indirect delivery always involves at least one:

A) Switch

B) Router

C) Repeater

D) Bridge

Section 2: 2 Mark Questions

1. Routing decides?

A) Best path

B) Frame size

C) Speed

D) Error

2. CRC is based on?

A) Polynomial division

B) Addition

C) Subtraction

D) Multiplication

3. Geostationary satellite completes orbit in?

A) 24 hours

B) 12 hours

C) 48 hours

D) 6 hours

4. Stop and Wait ARQ sends?

A) Random frames

B) Multiple frames

C) No frames

D) One frame at a time

5. IPv6 size?

A) 64 bits

B) 32 bits

C) 128 bits

D) 48 bits

6. ARP maps?

A) IP to MAC

B) MAC to IP

C) IP to Port

D) MAC to Port

7. IPv4 header is?

A) Static

B) Fixed

C) Random

D) Variable

8. Shortest path algorithm finds?

A) Fixed path

B) Longest path

C) Random path

D) Minimum cost path

9. Token bucket allows?

A) No traffic

B) Burst traffic

C) Constant only

D) Random

10. IPv6 header is?

A) Dynamic

B) Variable

C) Random

D) Fixed

Computer Networks - Set 3

Section 1: 1 Mark Questions

1. HTTP method used to request data without changes?

A) POST

B) PUT

C) GET

D) DELETE

2. In "Direct Delivery", source and destination are:

A) Different net

B) Same net

C) Multi-router

D) Diff protocol

3. Gathering data from different apps to the network layer is:

A) Demultiplexing

B) Segmentation

C) Multiplexing

D) Encapsulation

4. In circuit switching, a dedicated path is established:

A) After data transmission

B) Before data transmission

C) During data transmission

D) Only when error occurs

5. Primary advantage of Selective Repeat over GBN?

A) Simpler

B) Smaller buffer

C) Only damaged frame re-sent

D) No sequence #

6. Unlike TCP, SCTP uses a _________ handshake to establish associations.

A) Two-way

B) Three-way

C) Four-way

D) Five-way

7. In FTP Passive Mode, who initiates the data connection?

A) FTP Server

B) FTP Client

C) Firewall

D) Router

8. What does the sender do in Go-Back-N if a timer expires?

A) Retransmit one

B) Wait for request

C) Retransmit lost + subsequent

D) Drop connection

9. One major disadvantage of circuit switching is:

A) High delay in setup

B) Inefficient use of bandwidth

C) Packet loss

D) No dedicated path

10. What happens to a packet-switched network when the traffic load exceeds the network capacity?

A) The connection is dropped

B) Congestion and increased delay occur

C) The network speeds up

D) Data is automatically deleted

11. Which of the following is an example of circuit switching?

A) Internet

B) Email

C) Traditional telephone network

D) FTP

12. Mechanism to adjust data amount a sender can transmit?

A) Token Bucket

B) Sliding Window

C) Stop & Wait

D) Binary Backoff

13. A switched network is defined as a network:

A) With fixed paths only

B) That uses multiple possible paths for data transmission

C) Without routers

D) That uses only wireless communication

14. In Flooding, what happens when a packet arrives at a node?

A) Shortest path only

B) Sent to all except arrival line

C) Stored

D) Discarded

15. Which routing algorithm uses Dijkstra's algorithm for the shortest path?

A) Distance Vector

B) Link State

C) Flooding

D) Static

16. Which of the following is a key characteristic of the TCP protocol?

A) Connectionless

B) Unreliable delivery

C) Connection-oriented and reliable

D) No acknowledgments

17. Ports 0–1023 are known as:

A) Registered

B) Dynamic

C) Well-known

D) Private

18. In a Hamming code, at which positions are the redundant (parity) bits placed?

A) At the beginning

B) At the end

C) At positions that are powers of 2 (1, 2, 4, 8...)

D) Randomly

19. Which is an "Open Loop" congestion control policy?

A) Backpressure

B) Choke packets

C) Retransmission policy

D) Implicit signaling

20. Probability of transmission in 1-persistent CSMA if idle?

A) 0

B) 0.5

C) 1

D) 0.1

Section 2: 2 Mark Questions

1. Control signaling is used for?

A) Data transfer

B) Connection management

C) Encryption

D) Compression

2. Token bucket allows?

A) No traffic

B) Burst traffic

C) Constant only

D) Random

3. Routing decides?

A) Best path

B) Frame size

C) Speed

D) Error

4. CRC is based on?

A) Polynomial division

B) Addition

C) Subtraction

D) Multiplication

5. Checksum verifies?

A) Addressing

B) Routing

C) Data integrity

D) Timing

6. Stop and Wait ARQ sends?

A) Random frames

B) Multiple frames

C) No frames

D) One frame at a time

7. ALOHA is?

A) Routing protocol

B) Random access protocol

C) Error control

D) Switching

8. Parity bit is used for?

A) Compression

B) Routing

C) Encryption

D) Error detection

9. IPv4 header is?

A) Static

B) Fixed

C) Random

D) Variable

10. CSMA/CD is used in?

A) Ethernet

B) WiFi

C) Bluetooth

D) Satellite